mathongo

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- LATEST NTA PATTERN
- CONCEPT VIDEOS
- LIVE DOUBT SESSIONS
- DPP, REVISION SHEETS
- 100% MONEYBACK TO MERITORIOUS STUDENTS

JOIN NOW

YOUR LAST CHANCE!

Designed by
Math Maestro

Anup Sir

JEE Mains 2020 Jan Chapter wise Question Bank

Circle

01

Pair of tangents are drawn from origin to the circle $x^2 + y^2 - 8x - 4y + 16 = 0$ then square of length of chord of contact is

मूलबिन्दु से वृत्त $x^2 + y^2 - 8x - 4y + 16 = 0$ पर स्पर्श रेखाएें खीची जाती है तब स्पर्शजीवा की लम्बाई का वर्ग है–

(1) $\frac{64}{5}$ (2) $\frac{24}{5}$ (3) $\frac{8}{5}$ (4) $\frac{8}{13}$

7th Jan Evening

Sol

(1)

$L = \sqrt{S_1} = \sqrt{16} = 4$

$R = \sqrt{16+4-16} = 2$

Length of Chord of contact स्पर्शजीवा की लम्बाई $= \frac{2LR}{\sqrt{L^2+R^2}} = \frac{2\times4\times2}{\sqrt{16+4}} = \frac{16}{\sqrt{20}}$

square of length of chord of contact स्पर्शजीवा की लम्बाई का वर्ग $= \frac{64}{5}$

02

If $y = mx + c$ is a tangent to the circle $(x-3)^2 + y^2 = 1$ and also the perpendicular to the tangent to the circle $x^2 + y^2 = 1$ at $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, then

यदि रेखा $y = mx + c$ वृत $(x-3)^2 + y^2 = 1$ की स्पर्श रेखा है तथा वृत्त $x^2 + y^2 = 1$ के बिंदु $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ पर स्पर्श रेखा के लम्बवत् भी हैं तब

(1) $c^2 + 6c + 7 = 0$ (2) $c^2 - 6c + 7 = 0$ (3) $c^2 + 6c - 7 = 0$ (4) $c^2 - 6c - 7 = 0$

8th Jan Evening

Sol

(1)

Slope of tangent to $x^2 + y^2 = 1$ at $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$x^2 + y^2 = 1$ की $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ पर स्पर्श रेखा की प्रवणता

$x^2 + y^2 = 1$

$2x + 2yy' = 0$

$y' = -\frac{x}{y} = -1$

$y = mx + c$ is tangent of $x^2 + y^2 = 1$

$y = mx + c$ वृत्त $x^2 + y^2 = 1$ की स्पर्श रेखा है

so इसलिये $m = 1$

$y = x + c$

now distance of (3, 0) from $y = x + c$ is

अब (3, 0) की रेखा $y = x + c$ से दूरी है

$\left|\frac{c+3}{\sqrt{2}}\right| = 1$

$c^2 + 6c + 9 = 2$

$c^2 + 6c + 7 = 0$

03

If a circle touches y-axis at (0, 4) and passes through (2, 0) then which of the following can not be the tangent to the circle

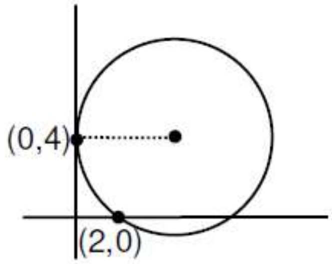
एक वृत्त y-अक्ष को (0, 4) पर स्पर्श करता है तथा (2, 0) से गुजरता है तो निम्न में से कौनसी रेखा वृत्त की स्पर्श रेखा नहीं है–

(1) $3x + 4y - 6 = 0$ (2) $3x + 4y - 24 = 0$ (3) $4x - 3y - 17 = 0$ (4) $4x + 3y - 6 = 0$

9^{th} Jan Morning

Sol

(1)



equation of family of circle वृत्त निकाय का समीकरण

$(x - 0)^2 + (y - 4)^2 + \lambda x = 0$

$\Rightarrow$ passes गुजरता है (2, 0)

$4 + 16 + 2\lambda = 0 \Rightarrow \lambda = -10$

$x^2 + y^2 - 10x - 8y + 16 = 0$

centre केन्द्र (5, 4). $R = \sqrt{25+16-16} = 5$

Check the options. विकल्पों की जाँच करें।

04

Let circles $(x - 0)^2 + (y - 4)^2 = k$ and $(x - 3)^2 + (y - 0)^2 = 1^2$ touches each other than find the maximum value of 'k'

9th Jan Evening

Sol

36.00

Two circles touches each other if $C_1 C_2 = |r_1 \pm r_2|$

Distance between $C_2(3, 0)$ and $C_1(0, 4)$ is either $\sqrt{k}+1$ or $\left|\sqrt{k}-1\right|$ $(C_1 C_2 = 5)$

$\Rightarrow \sqrt{k}+1 = 5$ or $\left|\sqrt{k}-1\right| = 5$ $\Rightarrow k = 16$ or $k = 36$ $\Rightarrow$ maximum value of k is 36